

When Does Adaptive Coordinate Sampling Pay for Itself?

Ridge-Regression Baselines, ASCD Oracle Quality, and Confirmatory Validation

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September 2026

Abstract

We study coordinate-selection rules for ridge regression under feature scaling and correlation. An initial frozen benchmark compares uniform sampling, fixed Lipschitz sampling, periodically refreshed gradient-gain sampling, and a progress-triggered gradient-gain heuristic across four synthetic regimes and the packaged diabetes regression dataset.

We subsequently implement Approximate Steepest Coordinate Descent (ASCD) of Stich, Raj, and Jaggi using the paper’s zero oracle g^3 with exact full-gradient initialization while retaining the benchmark’s exact ridge coordinate-minimization update. On the original scaled-correlated case, this configuration reaches the target in 9 of 12 runs compared with 6 of 12 for uniform sampling, but records greater proxy work on every seed where both methods succeed.

A Stage 4 oracle-quality analysis explains why zero-oracle ASCD provides little pruning. Its certified lower bounds collapse to zero on almost all coordinates, leaving an effectively full active set. Replacing the zero oracle by the exact inner-product oracle g^1 reduces the median active set to one coordinate. The tested simulated intermediate approximate oracles do not provide a useful middle regime.

A predeclared Stage 5 confirmatory experiment then evaluates uniform, fixed Lipschitz, g^3 , and g^1 on four held-out Toeplitz synthetic families and four additional real regression datasets, using a dimension-normalized 200-coordinate-epoch budget, a primary normalized objective-gap target of 10^{-6} , harmonized timing boundaries, and three single-threaded timing repetitions. The Stage 4 pruning mechanism generalizes to all eight problem families: the median g^3 active-set fraction is one and its median zero-lower-bound fraction is one, whereas g^1 has a one-coordinate median active set. Exact information has lower paired median time to target on five families but is substantially slower on both scaled synthetic families; on the strongly scaled correlated family, g^1 also reaches the primary target in only 9 of 12 runs compared with 12 of 12 for g^3 . None of the four methods reaches the primary target on Residential Building within the declared budget. These results support a mechanism-level conclusion about oracle quality and pruning, but not a claim of universal computational superiority or algorithmic novelty.

1 Motivation and related work

Randomized updates and adaptive information collection occur throughout modern optimization, including stochastic block BFGS [1], minibatch semi-stochastic optimization [2], parallel and accelerated coordinate descent [3, 5], and randomized iterative linear solvers [4]. Adaptive coordinate-selection methods include adaptive probabilities for stochastic dual coordinate ascent [6], adaptive importance sampling [7], and Approximate Steepest Coordinate Descent (ASCD) [8].

These works prevent gradient-informed coordinate selection or refresh rules alone from being treated as new ideas. The practical question in this project is instead:

When does additional information about coordinate quality improve optimization enough to compensate for the cost of obtaining, maintaining, and using that information?

The study is developed in four empirical layers.

First, a frozen baseline compares simple fixed and adaptive coordinate-sampling rules. Second, the published ASCD selection mechanism is implemented with the paper’s zero oracle on the same benchmark. Third, the oracle-quality mechanism is diagnosed by comparing zero, approximate, and exact gradient oracles. Fourth, a predeclared confirmatory experiment tests whether the resulting conclusions persist on held-out synthetic structures and additional real regression datasets under harmonized wall-clock accounting.

The later experiments are comparisons under this project’s ridge benchmark. They are not reproductions of the published ASCD paper’s figures.

2 Experimental setup and baseline methods

For a design matrix $X \in \mathbb{R}^{n \times d}$ and response $y \in \mathbb{R}^n$, we minimize

$$f(w) = \frac{1}{2} \|Xw - y\|_2^2 + \frac{\lambda}{2} \|w\|_2^2, \quad \lambda = 0.03. \quad (1)$$

With residual

$$r = Xw - y,$$

the coordinate gradient and coordinate smoothness are

$$g_j = X_j^\top r + \lambda w_j, \quad L_j = \|X_j\|_2^2 + \lambda. \quad (2)$$

Every method uses the same exact coordinate-minimization update,

$$w_j \leftarrow w_j - \frac{g_j}{L_j}, \quad (3)$$

with the residual updated incrementally.

Uniform sampling uses

$$p_j = \frac{1}{d},$$

whereas fixed Lipschitz sampling uses

$$p_j = \frac{L_j}{\sum_k L_k}.$$

The two baseline gradient-gain variants use

$$p_j = \frac{0.05}{d} + 0.95 \frac{g_j^2 / L_j}{\sum_k g_k^2 / L_k}, \quad (4)$$

with uniform fallback when the denominator vanishes.

Scheduled gain refreshes initially and every 60 updates. Triggered gain refreshes at step zero and subsequently at a 120-step checkpoint if the latest objective decrease is below half the preceding decrease, subject to a 60-step minimum interval.

The initial frozen benchmark contains four synthetic cases crossing balanced versus geometrically scaled features with independent versus eight-group correlated features. The synthetic cases use $n = 400$ and $d = 48$. The packaged diabetes regression design provides one additional real-data case.

Each baseline method uses 12 problem seeds, initializes at $w = 0$, and runs for at most 2,400 coordinate updates. A dense ridge solve defines f^* for evaluation only. A baseline run succeeds if a saved checkpoint satisfies

$$f(w) - f^* \leq 10^{-4}.$$

For the baseline study, the workload proxy counts modelled coordinate update, sampling, refresh, and evaluation operations. These units are bookkeeping quantities rather than measured floating-point-operation counts.

3 Baseline results

Table 1 reports the original baseline success counts.

Table 1: Baseline runs reaching the target within the 2,400-update budget, out of 12.

Case	Uniform	Lipschitz	Scheduled	Triggered
Balanced independent	12	12	12	12
Scaled independent	12	0	12	12
Balanced correlated	12	12	0	0
Scaled correlated	6	0	0	0
Diabetes	12	12	9	5

Uniform and fixed Lipschitz sampling succeed in all runs of the balanced-independent, balanced-correlated, and diabetes cases. Lipschitz sampling fails on the scaled-independent case, where the other three baseline methods succeed. Both gradient-gain heuristics fail to reach the target on either correlated synthetic design within the declared budget.

The baseline therefore does not provide evidence that the simple gradient-gain refresh mechanisms offer a consistent advantage.

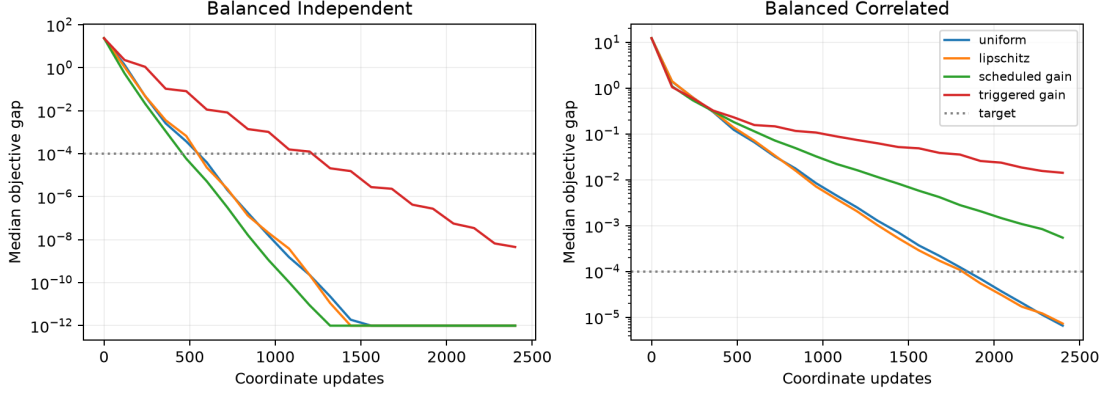


Figure 1: Median objective-gap trajectories for the balanced baseline cases. The horizontal target is 10^{-4} . Median trajectories summarize central behavior and do not show the full variation across individual runs.

4 Published ASCD comparison

4.1 Selection rule

ASCD [8] maintains, for each coordinate j , a stored gradient estimate $\tilde{g}_j$ and certified radius r_j satisfying

$$|g_j - \tilde{g}_j| \leq r_j. \quad (5)$$

The corresponding upper and lower magnitude bounds are

$$u_j = |\tilde{g}_j| + r_j, \quad \ell_j = \max\{0, |\tilde{g}_j| - r_j\}. \quad (6)$$

The active set is the smallest nonempty set I such that every excluded coordinate satisfies

$$u_j^2 < \frac{1}{|I|} \sum_{i \in I} \ell_i^2. \quad (7)$$

The exclusion inequality is strict; equality therefore retains the coordinate. Selection is uniform among active coordinates attaining the largest lower bound.

Our Stage 3 implementation initializes $\tilde{g}$ with the exact full gradient and initializes all radii to zero.

4.2 Zero oracle

For Stage 3, the paper’s zero cross-coordinate oracle g^3 is used. If coordinate i changes by h , then for $j \neq i$,

$$g_j^{\text{new}} - g_j = h X_j^\top X_i. \quad (8)$$

Cauchy–Schwarz gives

$$|g_j^{\text{new}} - g_j| \leq |h| \|X_i\|_2 \|X_j\|_2. \quad (9)$$

The stored passive-coordinate estimate is left unchanged and its radius is increased by this certified bound.

For the selected coordinate,

$$h = -\frac{g_i}{L_i},$$

so exact coordinate minimization gives

$$g_i^{\text{new}} = g_i + hL_i = 0$$

up to floating-point precision. Its stored estimate and radius can therefore be reset to zero.

4.3 Implementation verification

The implementation includes a numerical audit on a small dense ridge problem. After each audited update it recomputes

$$X^\top(Xw - y) + \lambda w$$

and verifies containment of every true coordinate gradient inside the maintained interval. The residual invariant

$$r = Xw - y$$

is also checked.

The strict active-set rule is tested separately, including its equality boundary, and the audited exact coordinate updates are required to be nonincreasing in objective value.

4.4 ASCD results

Stage 3 uses the same five frozen cases, 12 problem seeds, initial point, objective, 10^{-4} target, checkpoint spacing, and 2,400-update budget as the baseline.

Table 2: Uniform and zero-oracle ASCD results on the frozen benchmark. Work is the median recorded proxy work to target among successful runs.

Case	Uniform		ASCD g^3		
	Success	Work	Success	Work	Median $ I $
Balanced independent	12/12	513,288	12/12	762,936	48
Scaled independent	12/12	513,288	12/12	762,936	48
Balanced correlated	12/12	1,641,536	12/12	2,249,416	48
Scaled correlated	6/12	1,897,956	9/12	2,844,008	48
Diabetes	12/12	919,234	12/12	984,864	10

Zero-oracle ASCD therefore improves the success count on the scaled-correlated case from 6/12 to 9/12. However, the successful-run work medians in that row use different success sets and cannot by themselves establish a paired cost difference.

A later paired analysis identifies five scaled-correlated seeds on which both uniform and ASCD reach the target. On all five shared successes, ASCD records greater proxy work. The median paired recorded-work ratio

$$\frac{\text{ASCD work}}{\text{uniform work}}$$

is approximately 1.5401.

The active-set result is also important. The median active set is the entire coordinate dimension on every case: 48 coordinates on each synthetic problem and 10 coordinates on diabetes. Thus the zero-oracle mechanism provides almost no observed pruning under this configuration.

4.5 Stage 3 work accounting

The Stage 3 ASCD accounting includes an initial full-gradient charge of

$$nd + d.$$

Each saved objective checkpoint adds $n + d$. Each coordinate update adds

$$2n + 3 + d \lceil \log_2 d \rceil + 3d,$$

representing the coordinate update, modelled active-set computation, and interval maintenance.

These are approximate bookkeeping units rather than exact FLOP counts. Stage 3 baseline and ASCD timer boundaries were not fully harmonized, so the recorded wall times from that stage are not used as controlled method comparisons.

5 Stage 4: oracle-quality and active-set mechanism analysis

5.1 Why does the zero-oracle active set remain full?

Stage 4 first replays the frozen Stage 3 zero-oracle trajectory while recording per-update active-set and interval diagnostics. The replay reproduces the frozen Stage 3 objective checkpoints and proxy work.

For the four 48-coordinate synthetic cases, the full active set is used on approximately 99.94% of updates. On diabetes, the full 10-coordinate active set is used on approximately 99.96% of updates.

The median fraction of coordinates with zero certified lower bound is one on every case.

Since

$$\ell_j = \max\{0, |\tilde{g}_j| - r_j\},$$

a lower bound becomes zero whenever the certified radius is at least as large as the stored estimate magnitude. If all lower bounds are zero,

$$\frac{1}{|I|} \sum_{i \in I} \ell_i^2 = 0.$$

The strict exclusion condition in Equation (7) would then require

$$u_j^2 < 0,$$

which is impossible. The observed radius trajectories are consistent with this mechanism: zero-oracle uncertainty typically dominates the stored gradient information.

5.2 Oracle-quality experiment

Phase B retains the same frozen benchmark and exact coordinate update while varying the gradient oracle.

The configurations are

$$g^3, \quad g^2(\epsilon), \quad \epsilon \in \{1, 0.5, 0.25, 0.125\}, \quad g^1.$$

Here g^1 is the exact inner-product oracle. The simulated g^2 configurations use certified error

$$\delta_{ij} = \epsilon \|X_i\|_2 \|X_j\|_2.$$

The simulated g^2 experiment isolates oracle-quality effects. It does not model or charge a practical approximation-acquisition mechanism.

Table 3: Stage 4 oracle-quality runs reaching the target within 2,400 updates, out of 12.

Case	g^3	$g^2(1)$	$g^2(.5)$	$g^2(.25)$	$g^2(.125)$	g^1
Balanced independent	12	12	0	0	0	12
Scaled independent	12	12	0	0	0	12
Balanced correlated	12	12	0	0	0	12
Scaled correlated	9	11	0	0	0	12
Diabetes	12	12	0	0	0	12

5.3 Active-set transition

The exact oracle produces a qualitatively different active-set regime.

For g^1 , the median active set is one coordinate on every Stage 4 case, and the 10th, 25th, 50th, 75th, and 90th active-set percentiles are all one. By contrast, g^3 and $g^2(1)$ retain an effectively full active set.

The intermediate simulated g^2 configurations do prune frequently, but only weakly: the median synthetic active set is 47 of 48 coordinates and the diabetes median is 9 of 10 coordinates.

Thus the tested precision grid does not provide a useful intermediate regime between zero-oracle uncertainty and exact information.

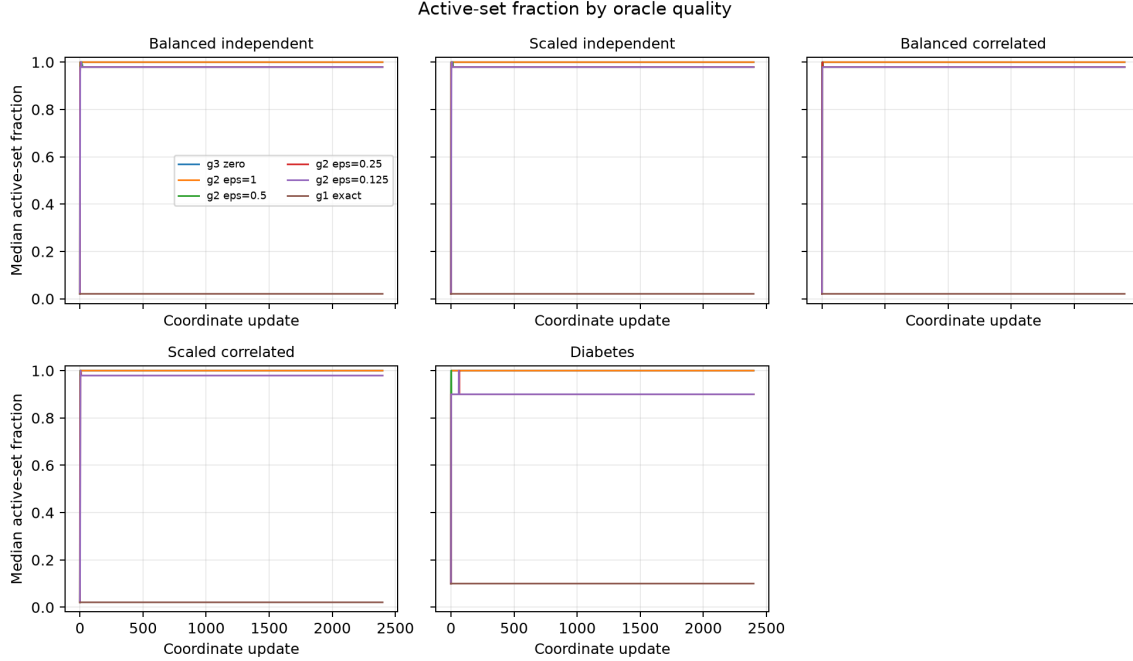


Figure 2: Stage 4 median active-set fraction across the oracle-quality configurations. Exact g^1 produces a one-coordinate active set, whereas the tested nonzero-error configurations remain close to the full coordinate set.

5.4 Optimization behavior

The objective-gap trajectories show a similarly sharp separation. Exact g^1 reaches the Stage 4 target on every run. The $\epsilon = 1$ simulated configuration largely resembles the zero oracle.

The tested $\epsilon \in \{0.5, 0.25, 0.125\}$ configurations do not reach the target on any Stage 4 run.

This nonmonotonic behavior should be interpreted narrowly. Changing the approximate oracle changes both the maintained estimates and the resulting coordinate-selection path. The result does not establish that a more accurate approximate oracle is generally harmful; rather, the tested simulated settings fail to produce a useful middle regime on this frozen benchmark.

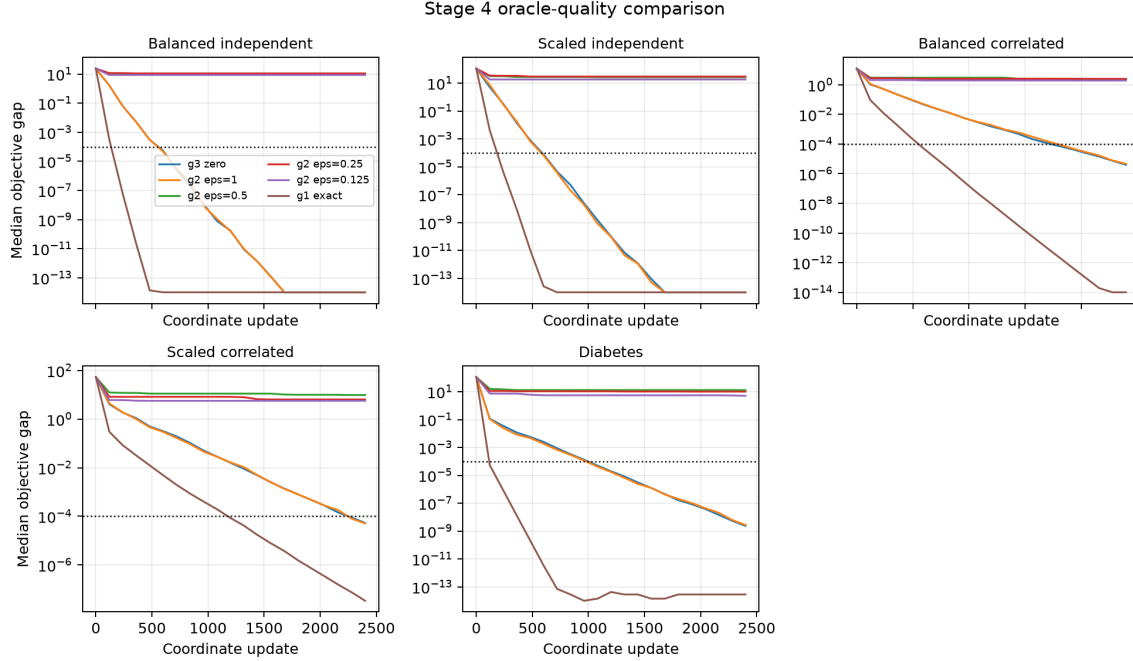


Figure 3: Stage 4 oracle-quality objective-gap trajectories. The exact oracle separates strongly from the zero oracle, whereas the tested approximate settings do not provide a successful intermediate regime.

5.5 Does exact information pay for itself under the Stage 4 proxy?

For g^1 , Stage 4 adds a dense Gram acquisition charge

$$nd^2$$

rather than treating exact inner products as free.

Table 4 reports the corresponding costed comparison.

Table 4: Stage 4 costed zero-oracle and exact-oracle comparison. The ratio is the median paired g^1/g^3 recorded-work ratio on shared successful runs.

Case	g^3 succ.	g^1 succ.	g^3 work	g^1 work	Paired ratio
Balanced independent	12/12	12/12	762,936	1,238,592	1.6235
Scaled independent	12/12	12/12	762,936	1,238,592	1.6235
Balanced correlated	12/12	12/12	2,249,416	1,684,536	0.7489
Scaled correlated	9/12	12/12	2,844,008	2,427,776	0.8536
Diabetes	12/12	12/12	984,864	164,374	0.1675

Exact information therefore does not have a uniform Stage 4 proxy-cost advantage. It is more expensive on the two independent synthetic cases, but cheaper on the correlated cases and diabetes under this accounting model.

These are proxy-accounting results, not controlled measured-runtime ratios.

6 Stage 5: confirmatory robustness and runtime validation

6.1 Predeclared held-out protocol

Stage 5 tests whether the Stage 4 mechanism persists outside the development benchmark.

The primary confirmatory question is:

Do the Stage 4 conclusions about oracle uncertainty, active-set pruning, and the cost of exact information persist on held-out synthetic structures and additional real regression datasets under harmonized runtime accounting?

Only four primary methods are retained:

- uniform coordinate sampling;
- fixed Lipschitz sampling;
- zero-oracle ASCD g^3 ;
- exact-inner-product ASCD g^1 .

The unsuccessful simulated Stage 4 g^2 precision grid is not included in the primary confirmatory study.

The held-out synthetic generator is deliberately different from the development generator. It uses

$$n = 800, \quad d = 96,$$

with Toeplitz covariance

$$\Sigma_{ij} = \rho^{|i-j|}, \quad \rho \in \{0.6, 0.9\}. \quad (10)$$

Balanced and geometrically scaled versions of each correlation level produce four held-out synthetic families. Scaled designs use

$$\text{logspace}(-1.5, 1.5, d).$$

Twelve problem seeds, 100 through 111, are used.

Four additional real regression datasets are included: California Housing, Concrete Compressive Strength, Airfoil Self-Noise, and Residential Building. After the fixed 80/20 split and training-only standardization, the corresponding training dimensions are

$$16512 \times 8, \quad 824 \times 8, \quad 1202 \times 5, \quad 297 \times 107.$$

For Residential Building, the sale-price output $V-9$ is the response; the second output $V-10$ is excluded from the design. Repeated economic variables are identified by their five source time lags.

Because Stage 5 spans substantially different sample sizes and scales, the primary convergence measure is the normalized objective gap

$$R_t = \frac{f(w_t) - f^*}{\max\{f(w_0) - f^*, 10^{-30}\}}. \quad (11)$$

The primary target is

$$R_t \leq 10^{-6},$$

and the secondary target is

$$R_t \leq 10^{-4}.$$

The maximum optimization budget is

$$200d$$

coordinate updates, i.e. 200 coordinate epochs, with checkpoints every $5d$ updates.

For timing, each method/problem pair receives one untimed warm-up and three timed repetitions. The timing boundary begins before method-specific setup and includes coordinate selection, updates, internal ASCD maintenance, and the exact Gram acquisition for g^1 . Checkpoint evaluation and the dense evaluation optimum are excluded from the primary method timer. Numerical kernels are constrained to one BLAS thread.

6.2 Primary success results

Table 5 reports the primary-target success counts.

Table 5: Stage 5 runs reaching the primary normalized-gap target $R_t \leq 10^{-6}$ within 200 coordinate epochs, out of 12.

Problem family	Uniform	Lipschitz	g^3	g^1
Balanced, $\rho = 0.6$	12	12	12	12
Balanced, $\rho = 0.9$	12	12	12	12
Scaled, $\rho = 0.6$	12	0	12	12
Scaled, $\rho = 0.9$	12	0	12	9
California Housing	12	12	12	12
Concrete Strength	12	12	12	12
Airfoil Self-Noise	12	12	12	12
Residential Building	0	0	0	0

Several results are important.

On both balanced held-out synthetic families, all four methods reach the primary target.

On both scaled held-out families, fixed Lipschitz sampling fails in all 12 runs. Uniform and zero-oracle ASCD reach the target in every run. Exact g^1 reaches it in all 12 moderate-correlation scaled runs but in only 9 of 12 strongly correlated scaled runs.

On California Housing, Concrete, and Airfoil, every method reaches the primary target in all 12 algorithmic runs.

Residential Building is a useful negative case: no method reaches the primary 10^{-6} target within the declared budget. Exact g^1 does, however, reach the secondary 10^{-4} target in all 12 runs, with a median final normalized gap of approximately 7.45×10^{-5} .

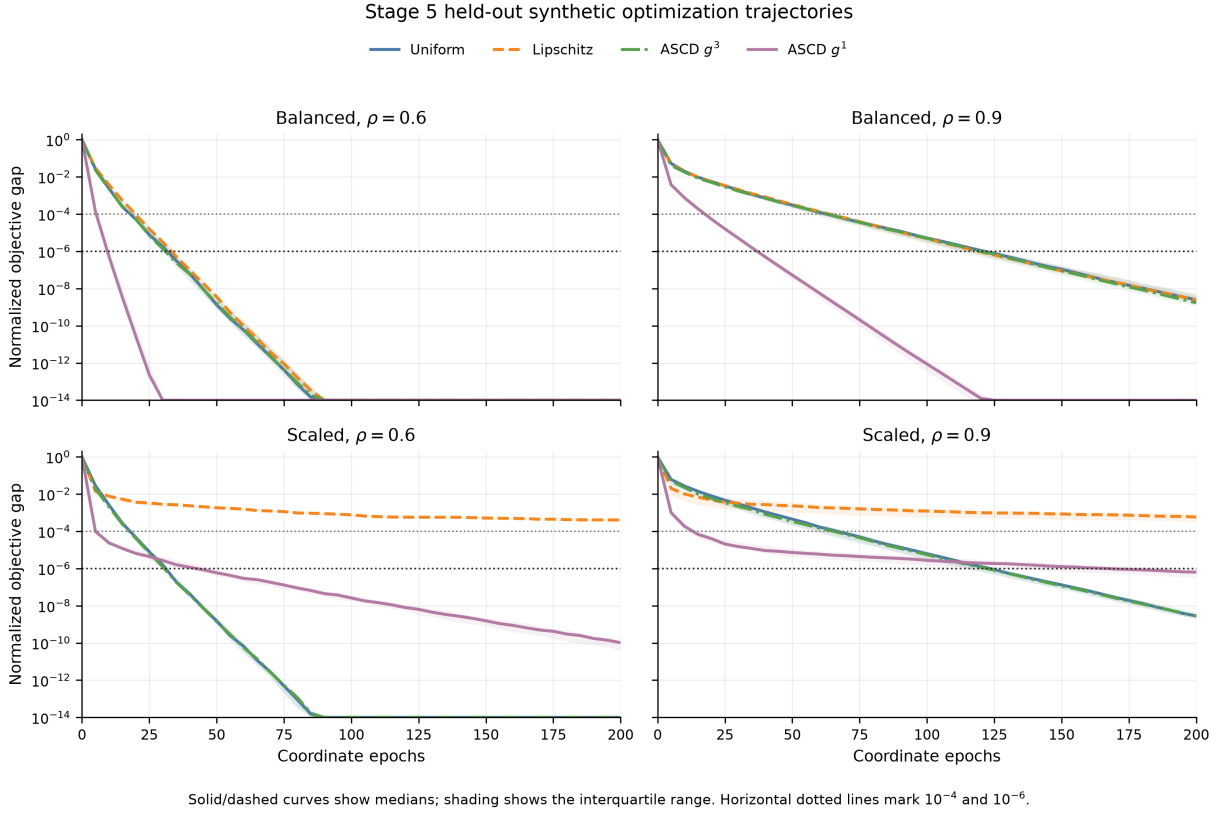


Figure 4: Stage 5 held-out synthetic normalized objective-gap trajectories. Curves are medians across the 12 paired problem instances and shading is the interquartile range. The horizontal dotted lines mark the secondary 10^{-4} and primary 10^{-6} targets.

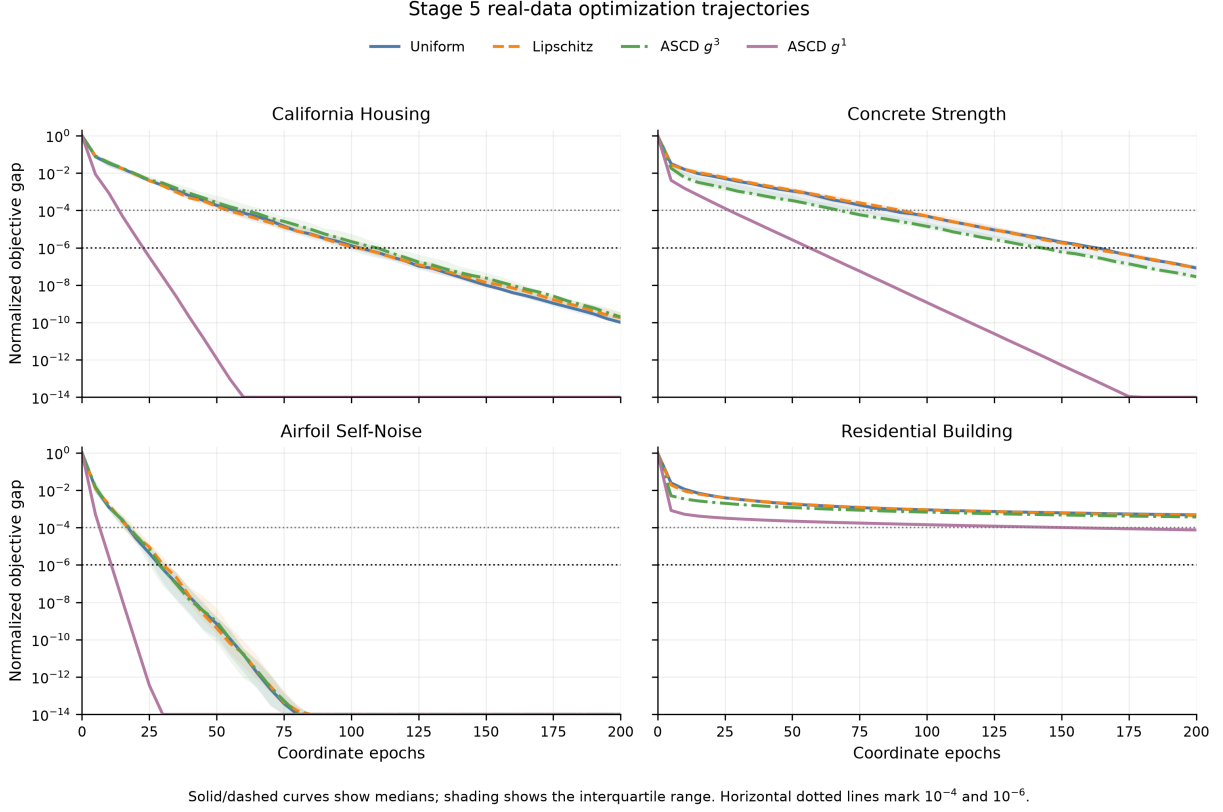


Figure 5: Stage 5 real-data normalized objective-gap trajectories. The fixed split is shared across methods, while the 12 repetitions use the predeclared algorithm seeds.

6.3 Does the Stage 4 active-set mechanism generalize?

The confirmatory mechanism result is unusually consistent.

For zero-oracle g^3 , the median active-set fraction is 1.0 on every one of the eight Stage 5 problem families. The median fraction of coordinates with zero certified lower bound is likewise 1.0 on all eight.

Exact g^1 , by contrast, has a median active-set size of one coordinate on every problem family. Consequently its median active-set fraction is

$$\frac{1}{96} \approx 0.0104$$

on the held-out synthetic families,

$$\frac{1}{8} = 0.125$$

on California and Concrete,

$$\frac{1}{5} = 0.2$$

on Airfoil, and

$$\frac{1}{107} \approx 0.00935$$

on Residential Building.

Table 6 summarizes this contrast.

Table 6: Stage 5 median ASCD active-set and zero-lower-bound fractions.

Problem family	Active-set fraction		Zero-lower fraction	
	g^3	g^1	g^3	g^1
Balanced, $\rho = 0.6$	1.000	0.0104	1.000	0.0104
Balanced, $\rho = 0.9$	1.000	0.0104	1.000	0.0104
Scaled, $\rho = 0.6$	1.000	0.0104	1.000	0.0104
Scaled, $\rho = 0.9$	1.000	0.0104	1.000	0.0104
California Housing	1.000	0.1250	1.000	0.1250
Concrete Strength	1.000	0.1250	1.000	0.1250
Airfoil Self-Noise	1.000	0.2000	1.000	0.2000
Residential Building	1.000	0.00935	1.000	0.00935

Thus the predeclared Stage 5 mechanism criterion is met on all eight problem families, rather than merely on the required majority.

The Stage 4 explanation therefore survives the held-out experiment: zero-oracle uncertainty repeatedly collapses the lower bounds and leaves the safe active set effectively unpruned, whereas exact oracle information produces one-coordinate selection.

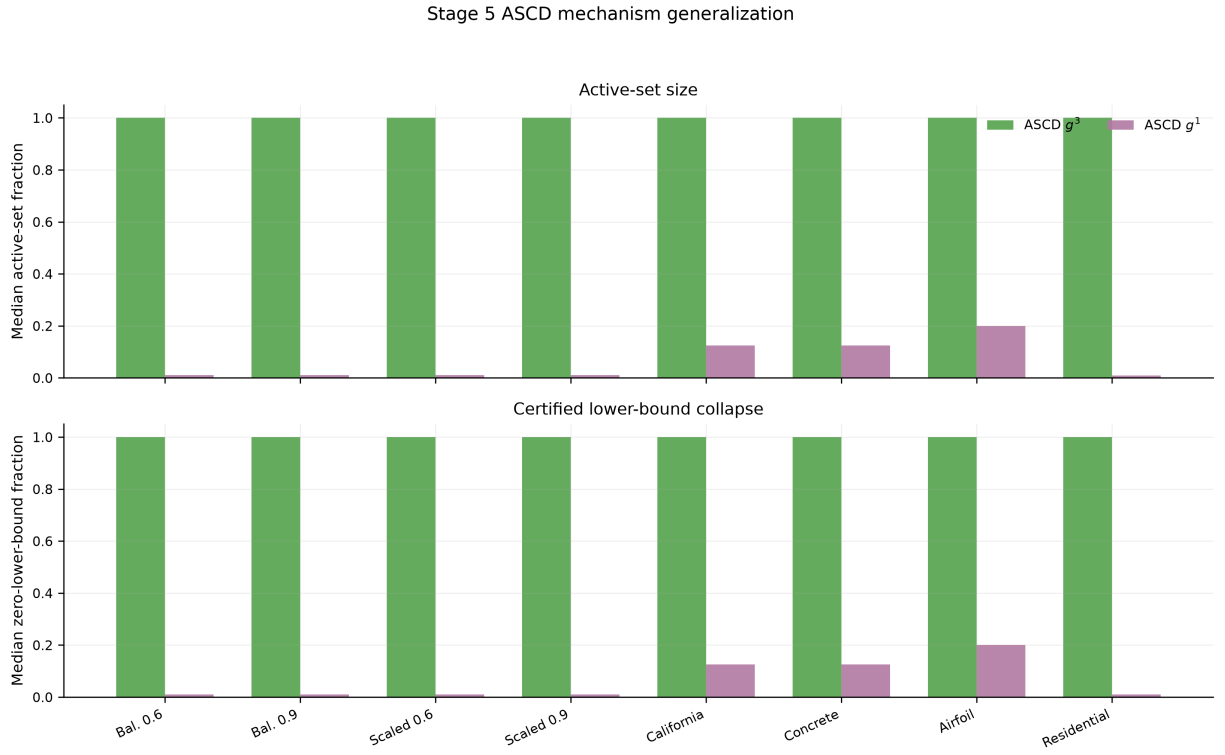


Figure 6: Stage 5 ASCD mechanism generalization. Zero-oracle g^3 retains the full active set and exhibits complete lower-bound collapse, whereas exact g^1 produces an effectively one-coordinate active set on every held-out family.

6.4 Coordinate effort and harmonized runtime

Table 7 compares exact and zero-oracle ASCD on their shared primary successes.

The epoch ratio is

$$\frac{g^1 \text{ coordinate epochs to target}}{g^3 \text{ coordinate epochs to target}},$$

and the runtime ratio uses the corresponding harmonized measured time. Reported runtime intervals are percentile 95% confidence intervals from 10,000 paired bootstrap resamples.

Table 7: Paired Stage 5 g^1/g^3 comparisons on shared primary successes. Runtime ratios below one favor g^1 under the measured implementation.

Problem family	g^3 succ.	g^1 succ.	Epoch ratio	Runtime ratio [95% CI]
Balanced, $\rho = 0.6$	12/12	12/12	0.3095	0.8179 [0.6954, 0.9811]
Balanced, $\rho = 0.9$	12/12	12/12	0.3191	0.8389 [0.7417, 0.9123]
Scaled, $\rho = 0.6$	12/12	12/12	1.4286	3.2393 [2.6883, 4.2613]
Scaled, $\rho = 0.9$	12/12	9/12	1.2593	3.1274 [2.1715, 3.3162]
California Housing	12/12	12/12	0.2273	0.2080 [0.1859, 0.2802]
Concrete Strength	12/12	12/12	0.4138	0.3779 [0.3655, 0.4520]
Airfoil Self-Noise	12/12	12/12	0.5000	0.5132 [0.3836, 0.6628]
Residential Building	0/12	0/12	–	–

The timing result is therefore mixed rather than monotone.

Exact information is faster than zero-oracle ASCD in paired median time to the primary target on the two balanced synthetic families and on California, Concrete, and Airfoil.

It is substantially slower on both scaled synthetic families. On the moderately correlated scaled family, the paired median runtime ratio is approximately 3.24. On the strongly correlated scaled family it is approximately 3.13 among the nine shared successes, and g^1 also fails on three instances on which g^3 succeeds.

The predeclared criterion that exact information can *sometimes* pay for itself is therefore met, but the stronger claim that exact information is generally preferable is not supported.

This Stage 5 result also improves on the earlier accounting-only comparison because the runtime boundary is harmonized and the exact Gram construction is inside the g^1 timer.

6.5 Real-data predictive diagnostics

Stage 5 remains an optimization experiment rather than a predictive machine-learning model-selection study.

All four methods optimize the same training ridge objective. Test RMSE is therefore recorded only as a secondary diagnostic of the final iterates.

On California, Concrete, and Airfoil, the methods that converge tightly to the ridge optimum also produce test RMSE values numerically very close to the dense reference solution.

Residential Building illustrates why these RMSE values should not be used to rank the optimization methods. None of the methods reaches the primary optimization target, and some incompletely optimized iterates obtain lower test RMSE than the exact training optimum. For example, the median final g^3 test RMSE is below the dense ridge reference test RMSE even though g^3 has a substantially larger training objective gap. This is possible because minimizing the fixed training objective is not equivalent to minimizing test RMSE.

Accordingly, Stage 5 does not interpret test RMSE differences as an optimization-method superiority result.

6.6 Predeclared confirmatory decisions

The Stage 5 protocol specified four mechanism and cost questions.

The resulting decisions are as follows.

1. **Zero-oracle pruning failure.** The median g^3 active-set fraction exceeds the g^1 fraction on all eight problem families, exceeding the predeclared majority requirement.
2. **Exact-information pruning.** Exact g^1 reduces the median active set to one coordinate on every family. It also uses fewer paired median coordinate epochs than g^3 on the two balanced synthetic families and on California, Concrete, and Airfoil.
3. **Regime-dependent information cost.** The paired median runtime ratio is below one on five families and above one on both scaled synthetic families. Residential has no shared primary success and therefore no time-to-target ratio.
4. **Generalization of the Stage 4 mechanism.** Both the active-set and zero-lower-bound criteria hold on all eight held-out families.

The predeclared rule that exact information can sometimes pay for itself is therefore satisfied. The confirmatory evidence does not support a universal exact-oracle advantage.

7 Interpretation and limitations

The combined experiments support a narrower and more defensible conclusion than a general claim that adaptive coordinate selection is better.

The baseline experiment demonstrates that simple gradient-gain refresh heuristics are highly sensitive to problem structure and do not consistently improve on fixed sampling.

Stage 3 shows that a published ASCD selection mechanism can improve success count on one difficult development regime, but that the zero oracle retains a full active set and can incur greater recorded work.

Stage 4 identifies the mechanism behind that behavior. The Cauchy–Schwarz zero-oracle radii are sufficiently broad that certified lower bounds collapse to zero on almost all coordinates. The safe exclusion test then has essentially no opportunity to prune. Exact inner-product information removes that uncertainty and produces a one-coordinate active set.

Stage 5 provides the strongest evidence for this mechanism because the same qualitative effect appears on all four held-out Toeplitz synthetic families and all four additional real datasets. The mechanism is therefore not confined to the original grouped-correlation synthetic generator or the diabetes dataset.

However, aggressive pruning is not equivalent to universal computational advantage. On two balanced synthetic families and three real datasets, the reduction in coordinate effort is large enough that g^1 has a lower measured time to the primary target than g^3 . On both scaled synthetic families the direction reverses. In the strongest scaled case, g^1 also has a lower primary success count.

Several limitations remain.

First, the experiments concern convex ridge regression with exact coordinate minimization. They do not establish the same behavior for general smooth objectives, nonconvex machine-learning models, or high-dimensional sparse learning problems.

Second, the Stage 5 real datasets broaden the evaluation substantially relative to the original diabetes-only benchmark, but four additional regression datasets are still a limited empirical sample.

Third, measured runtimes remain implementation- and hardware-dependent. Stage 5 improves internal validity through harmonized timing, single-threaded numerical kernels, warm-up execution, and repeated measurement, but its absolute timing ratios should not be assumed to transfer unchanged to other implementations.

Fourth, proxy work remains a modelled accounting quantity rather than an exact floating-point-operation count.

Fifth, the exact oracle obtains its information through a dense Gram matrix. This is a useful controlled reference point but may be impractical for genuinely large d . The tested Stage 4 simulated approximate oracles do not establish a practical approximate alternative.

Sixth, the theoretical one-step comparison in Stich et al. should not be treated as a direct guarantee for this benchmark. This project retains coordinate-specific L_j values and exact coordinate minimization, so its experimental update convention differs from assumptions required for a direct application of the paper’s comparison theorem.

Finally, objective gaps reported at or near zero can be limited by floating-point precision. All convergence claims should therefore be understood relative to the declared numerical targets and implementation precision.

8 Conclusion and future research

This study began with a simple practical question: when does the additional information used by adaptive coordinate selection justify its computational cost?

The baseline study shows that the answer cannot be reduced to “adaptive is better.” The tested gradient-gain refresh heuristics succeed on some regimes and fail on others.

The published zero-oracle ASCD comparison likewise gives a mixed result. It increases scaled-correlated success count on the original benchmark but provides almost no active-set pruning and records greater paired proxy work than uniform sampling on their shared successful instances.

The Stage 4 oracle analysis identifies a concrete mechanism for this failure to prune. Under g^3 , certified uncertainty drives the lower bounds to zero and leaves the safe active set effectively full. Exact g^1 reverses this behavior and reduces selection to one coordinate.

Most importantly, Stage 5 confirms that this mechanism persists beyond the development benchmark. On all eight held-out problem families, zero-oracle ASCD retains a full median active set with complete median lower-bound collapse, whereas exact-oracle ASCD has a one-coordinate median active set.

The computational consequence is regime-dependent. Exact information has lower paired median time to target on five Stage 5 families but is more than three times as slow as g^3 on the two scaled synthetic families under the measured implementation. On the strongest scaled family it also succeeds less often at the primary target. Residential Building supplies a separate negative result in which no method reaches the primary target within the declared budget.

The resulting research conclusion is therefore not that exact information is always worth acquiring. Rather, *oracle information quality strongly determines whether the ASCD active-set mechanism can prune, but whether that information pays for itself depends on the problem regime and information-acquisition cost.*

A publication-oriented extension could investigate a genuinely costed approximate-inner-product oracle or another theoretically motivated information-acquisition rule. Such an extension should define the acquisition mechanism and its cost before inspecting results and should be evaluated separately from the completed confirmatory study.

Before any such methodological extension, the present work can be completed through a final

reproducibility and synthesis stage: auditing all reported numbers against the generated artifacts, freezing the experimental outputs, pinning the computational environment, documenting exact reproduction commands, and performing a final page-by-page report review.

Acknowledgement and research provenance. AI assistance was used in developing the baseline benchmark, ASCD implementation, Stage 3 paired analysis, Stage 4 diagnostic and oracle-comparison workflows, Stage 5 protocol and implementation, plotting, analysis, and report text. The code, protocol documents, and generated outputs accompany the report for reproducibility. Descriptions of independent authorship should reflect the author’s actual personal verification, decisions, and revisions. Applicable citation and AI-disclosure requirements should be followed.

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